Notes on Van der Corput's Resolution

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Contents

1 Theorem 2.2 of Van der Corput

Theorem 1.1. *Assume f is twice differentiable on $I = [a, b]$, and there exist constants $\lambda > 0$ and $\alpha \geqslant 1$ such that*

$$\lambda \leqslant |f''(x)| \leqslant \alpha\lambda \qquad (x \in I).$$

Then

$$\sum_{n \in I} \mathrm{e}(f(n)) \ll \alpha\, |I|\, \lambda^{1/2} + \lambda^{-1/2}.$$

Proof. Let $\mathrm{e}(t) = e^{2\pi i t}$, denote $|I| = b - a$, and let $\|x\|$ be the distance from x to the nearest integer.

Since f'' does not change sign on I, f' is strictly monotonic. By the mean value theorem,

$$|f'(b) - f'(a)| \leqslant \alpha\lambda\, |I|\,. \tag{1}$$

Splitting the interval

Fix $0 < \delta < \frac{1}{2}$. Define the sets

$$G = \{x \in I : \|f'(x)\| \geqslant \delta\} \quad \text{(good set)}, \qquad B = I \setminus G \quad \text{(bad set)}.$$

Since f' is monotone, each component of B is of the form

$$J_m := f'^{-1}\big([m - \delta,\, m + \delta]\big), \qquad m \in \mathbb{Z}.$$

Such intervals exist only if they intersect $f'(I) = [f'(a), f'(b)]$. Hence, the number of relevant m is bounded by

$$\#\{m : J_m \neq \varnothing\} \;\leqslant\; |f'(b) - f'(a)| + 2 \;\leqslant\; \alpha\lambda\, |I| + 2. \tag{2}$$

Therefore:

- The number of bad intervals is $\ll \alpha\lambda\, |I| + 1$.
- The number of good intervals is at most one more than the number of bad intervals, hence also $\ll \alpha\lambda\, |I| + 1$.

Moreover, for each bad interval J_m, the length is bounded by

$$|J_m| \leqslant \frac{2\delta}{\lambda}, \tag{3}$$

since $|f''| \geqslant \lambda$.

Estimates on intervals

For a set E, write

$$S(E) := \sum_{n \in E \cap \mathbb{Z}} \mathrm{e}(f(n))\,.$$

- **Good intervals ($\|f'\| \geqslant \delta$):** By the first derivative test (Kusmin–Landau lemma),

$$|S(J)| \;\ll\; \delta^{-1}, \qquad J \subset G. \tag{4}$$

- **Bad intervals ($\|f'\| < \delta$):** Trivially,

$$|S(J_m)| \;\leqslant\; \#(J_m \cap \mathbb{Z}) \;\ll\; |J_m| + 1 \overset{(3)}{\ll} \tfrac{\delta}{\lambda} + 1. \tag{5}$$

Global estimate

Combining (2), (4), and (5), we obtain

$$\sum_{n \in I} \mathrm{e}(f(n)) = \sum_{J \subset G} S(J) + \sum_{J_m \subset B} S(J_m) \;\ll\; \big(\alpha\lambda\, |I| + 1\big)\left(\tfrac{1}{\delta} + \tfrac{\delta}{\lambda} + 1\right). \tag{6}$$

Optimizing δ

Choose $\delta = \lambda^{1/2}$. If $\lambda \leqslant \frac{1}{4}$, then $\delta < \frac{1}{2}$, so (6) yields

$$\sum_{n \in I} \mathrm{e}(f(n)) \ll \alpha\, |I|\, \lambda^{1/2} + \lambda^{-1/2}.$$

If $\lambda > \frac{1}{4}$, the right-hand side is $\asymp \alpha\, |I|$. The trivial bound $\left|\sum_{n\in I} \mathrm{e}(f(n))\right| \leqslant |I| + 1$ gives the same order, so

$$\sum_{n \in I} \mathrm{e}(f(n)) \ll \alpha\, |I|\, \lambda^{1/2} + \lambda^{-1/2}.$$

This completes the proof. $\square$

2 Lemma 3.6 of Exponent Pairs

Lemma 2.1. *Suppose that f has four continuous derivatives on $[a, b]$, and that $f' < 0$ on this interval. Suppose further that $[a, b] \subseteq [N, 2N]$ and that $\alpha = f'(b)$ and $\beta = f'(a)$. Assume that there is some $F > 0$ such that*

$$f^{(2)}(x) \approx FN^{-2}, \quad f^{(3)}(x) < FN^{-3}, \quad \textit{and} \quad f^{(4)}(x) < FN^{-4}$$

for x in $[a, b]$. Let x_ν be defined by the relation $f'(x_\nu) = \nu$, and let $\phi(\nu) = -f(x_\nu) + \nu x_\nu$. Then

$$\sum_{n\in I} \mathrm{e}(f(n)) = \sum_{\alpha<\nu\leqslant\beta} \frac{\mathrm{e}(-\phi(\nu) - 1/8)}{|f''(x_\nu)|^{1/2}} + O(\log(FN^{-1} + 2) + F^{-1/2}N).$$

Proof. We may assume that $F \geqslant 1$, for the theorem is trivial otherwise. Let $H_1 = [\alpha] - 1$ and $H_2 = [\beta] + 1$. From the previous lemmas, we see that

$$\sum_{n\in I} \mathrm{e}(f(n)) = \sum_{H_1\leqslant\nu\leqslant H_2} \frac{\mathrm{e}(-\phi(\nu) - 1/8)}{|f''(x_\nu)|^{1/2}} + O(E),$$

where

$$E = \sum_{H_1\leqslant\nu\leqslant H_2} F^{-1}N + \min(|f'(a) - \nu|^{-1}, F^{-1/2}N) + \min(|f'(b) - \nu|^{-1}, F^{-1/2}N).$$

Note that

$$H_2 - H_1 < \left| \int_a^b f''(x)\, dx \right| + 1 < FN^{-1} + 1,$$

so

$$\sum_{H_1\leqslant\nu\leqslant H_2} F^{-1}N < F^{-1}N(FN^{-1} + 1) < 1 + F^{-1}N.$$

The second term is majorized by $F^{-1/2}N$ since we are assuming that $F \geqslant 1$. Note also that for $\nu < H_2 - 1$,

$$|f'(a) - \nu| \geqslant H_2 - 1 - \nu$$

so

$$\sum_{H_1\leqslant\nu\leqslant H_2} \min(|f'(a) - \nu|^{-1}, F^{-1/2}N) < F^{-1/2}N + \sum_{H_1\leqslant\nu<H_2-1} (H_2 - 1 - \nu)^{-1} < F^{-1/2}N + \log(FN^{-1} + 2).$$

The same upper bound holds for the sum

$$\sum_{H_1\leqslant\nu\leqslant H_2} \min(|f'(b) - \nu|^{-1}, F^{-1/2}N).$$

Finally, if we change the range of summation in (3.2.6) to $\alpha < \nu \leqslant \beta$, this changes the sum by $< F^{-1/2}N$, and this may be absorbed into the error term. $\square$

3 Heuristic Definitions

Definition 3.1. *Let $N, y, s, \varepsilon < \frac{1}{2} \in \mathbb{R}_+$, and $P \in \mathbb{Z}_+$. We define the set*

$$\begin{aligned}&\mathbf{F}(N,P,s,y,\varepsilon)\\ :=&\left\{ f \in C^P([a,b]) :\ \forall 0 \leqslant p \leqslant P-1, a \leqslant x \leqslant b,\ \left| f^{(p+1)}(x) - (-1)^p (s)_p y x^{-s-p} \right| < \varepsilon (s)_p y x^{-s-p} \right\},\end{aligned}$$

in which $[a,b] \subset [N, 2N]$, and $(s)_p = s(s+1)(s+2)\cdots(s+p-1) = \frac{(s+p-1)!}{(s-1)!}$. $f \in C^p(\Omega)$ means that f is defined and has p continuous derivatives on the set Ω.

The set $\mathbf{F}(N,P,s,y,\varepsilon)$ consists of all functions $f \in C^P([a,b])$ whose derivatives up to order P closely resemble those of the model phase $f_0(x) = -yx^{-s}$. Specifically, for each $0 \leqslant p \leqslant P-1$ and $x \in [a,b] \subset [N,2N]$,

$$f^{(p+1)}(x) = (-1)^p (s)_p y x^{-s-p} (1 + O(\varepsilon)),$$

or equivalently,

$$(1-\varepsilon)(s)_p y x^{-s-p} < (-1)^p f^{(p+1)}(x) < (1+\varepsilon)(s)_p y x^{-s-p}.$$

Thus, $\mathbf{F}(N,P,s,y,\varepsilon)$ captures all phase functions whose derivatives follow the same scaling pattern as x^{-s-p}, up to a small relative error ε.

Definition 3.2. *Let $k, l \in \mathbb{R}$ such that $0 \leqslant k \leqslant \frac{1}{2} \leqslant l \leqslant 1$. $\forall s > 0,\ \exists P = P(k,l,s), \varepsilon = \varepsilon(k,l,s)$ s.t. $\forall N > 0, y > 0,\ f \in \mathbf{F}(N,P,s,y,\varepsilon)$, the estimate*

$$\sum_{n \in I} \mathrm{e}(f(n)) \ll L^k N^l + L^{-1}$$

*where $L = yN^{-s}$. Then, (k,l) is an **exponent pair**.*

The combination of $L^k N^l$ and L^{-1} is due to the fact that for cases $L \gg 1$, the $L^k N^l$ dominates, and L^{-1} dominates if $L \ll 1$. Since we usually assume that $L \gg 1$, we can reduce the estimate to $S \ll L^k N^l$.

We are able to summarize the standardized method consisting of Weyl differencing and Cauchy-Schwartz in the equidistribution problems we resolved. By observing the change of parameters when we conduct one differencing step, we summarized Table 1. To sum up, we introduce the Van der Corput A-process, the normalized, standardized resolution.

Theorem 3.3 (Van der Corput **A-process**)**.** *If (k,l) is an exponent pair, then*

$$A(k,l) := \left(\frac{k}{2k+2}, \frac{k+l+1}{2k+2} \right)$$

is also an exponent pair.

From now on, the estimation can be attained mechanically.

Param.	Original $f(x)$	After differencing $f_1(x)=f(x+h)-f(x)$	Change	Explanation
$\mathbf{F}$	$f\in\mathbf{F}(N,P,s,y,\varepsilon)$	$f_1 \in \mathbf{F}(N,P-1,s+1,shy,3\varepsilon)$	$P\to P-1$, $s\to s+1$, $y\to shy$	Differencing lowers smoothness (P), raises degree (s), scales y by sh.
L	$L=yN^{-s}$	$L_1=(shy)N^{-(s+1)}=shLN^{-1}$	$L\to shLN^{-1}$	L shrinks by N^{-1} and multiplies by sh.
Effective L	L	$L'=hLN^{-1}$	$L'\approx L_1/s$	s is fixed, so absorbed into the implied constant.
N	N	N (unchanged)	—	Differencing acts on f, not on the summation interval.
s	Fixed degree parameter	New fixed value $s+1$	$s\to s+1$	s treated as constant within each step.
P	P	$P-1$	$P\to P-1$	Smoothness order decreases by one.
ε	ε	3ε	$\varepsilon\to 3\varepsilon$	Error term grows after differencing.
y	y	shy	$y\to shy$	From $f(x+h)-f(x)\approx f'(x)h\sim syhx^{s-1}$.

Table 1: Parameter transitions after one differencing step in van der Corput's method.

4 The Proof of B-process

Lemma 4.1. *For function $f\in\mathbf{F}(N,F,s,y,\varepsilon)$, we define the $\sigma=\frac{1}{s}$, and $\eta=y^{\sigma}$; assume that f is defined on $[a,b]$ and $\nu\in[\alpha,\beta]=[f'(b),f'(a)]$. x_ν satisfies $f'(x_\nu)=\nu$. The auxiliary function $\phi(\nu)=\nu x_\nu-f(x_\nu)$. Then, $\exists\, C=C(s,P)$ constant that*

$$\left|\phi^{(p+1)}(\nu)-(-1)^p(s)_p\eta\nu^{-\sigma-p}\right|<C\varepsilon(s)_p\eta\nu^{-\sigma-p}.$$

Proof. We first incorporate the bound function F:

$$F(x):=\begin{cases} y\frac{x^{1-s}}{1-s} & \text{for} \quad s\neq 1,\\ y\log x & \text{for} \quad s=1\end{cases}$$

Then the condition $f\in\mathbf{F}(N,F,s,y,\varepsilon)\iff\left|f^{(p+1)}(x)-F^{(p+1)}(x)\right|<\varepsilon\left|F^{(p+1)}(x)\right|$. From this perspective, we can also create a function that represents the bound of $\phi(\nu)$.

The functional within the bound

For X_ν satisfies $F'(X_\nu)=\nu$ and $\Phi(\nu)=\nu X_\nu-F(X_\nu)$, we can actually deduce that $X_\nu=\eta\nu^{-\sigma}$. Then, the lemma is reduced to

$$\left|\phi^{(p+1)}(\nu)-\Phi^{(p+1)}(\nu)\right|<C\varepsilon\left|\Phi^{(p+1)}(\nu)\right| \tag{7}$$

We recognize that Phi is a sort of functional.

By Chain Rule, we derived that the $\phi''(\nu)=\frac{1}{f''(x_\nu)}$ from $f'(\phi'(\nu))=\nu$. If we iterate,

$$\begin{aligned}\phi^{(3)}(\nu)&=\frac{1}{(f''(x_\nu))^3}(-1)f^{(3)}(x_\nu)\\ \phi^{(4)}(\nu)&=\frac{1}{(f''(x_\nu))^4}(f^{(3)}(x_\nu)-f^{(4)}(x_\nu))\\ &\cdots\\ \phi^{(p+1)}(\nu)&=\frac{1}{(f''(x_\nu))^{2p-1}}\sum_{\mathbf{u}}w_{\mathbf{u}}\prod_{i=1}^{p-1}f^{(u_i+1)}(x_\nu)\end{aligned}$$

and we found that the $\sum_i u_i=2p-2$; $w_{\mathbf{u}}$ is a constant that depends only on u_i. Similarly,

$$\begin{cases}\phi^{(p+1)}(\nu)=\frac{1}{(f''(x_\nu))^{2p-1}}\sum_{\mathbf{u}}w_{\mathbf{u}}\prod_{i=1}^{p-1}f^{(u_i+1)}(x_\nu)\\ \Phi^{(p+1)}(\nu)=\frac{1}{(F''(X_\nu))^{2p-1}}\sum_{\mathbf{u}}w_{\mathbf{u}}\prod_{i=1}^{p-1}F^{(u_i+1)}(X_\nu)\end{cases} \tag{8}$$

Thus, we reveal the structure of the functional Phi: $\varphi^{(p+1)}(f) = \frac{1}{(f''(x_\nu))^{2p-1}} \sum_{\mathbf{u}} w_{\mathbf{u}} \prod_{i=1}^{p-1} f^{(u_i+1)}(x_\nu)$

From function difference to functional difference

We have to estimate the difference $\left|f^{(p+1)}(x_\nu) - F^{(p+1)}(X_\nu)\right| < \varepsilon \left|F^{(p+1)}(X_\nu)\right|$ so that we are able to get

$$f^{(p+1)}(x_\nu) \asymp_\varepsilon F^{(p+1)}(x_\nu).$$

Since we already have $\left|f^{(p+1)}(x) - F^{(p+1)}(x)\right| < \varepsilon \left|F^{(p+1)}(x)\right|$, what's left is

$$\left|F^{(p+1)}(x_\nu) - F^{(p+1)}(X_\nu)\right| < \varepsilon \left|F^{(p+1)}(X_\nu)\right|.$$

We can apply the Mean Value Theorem: $\exists\, t$ between endpoints x_ν, X_ν such that

$$\left|F^{(p+1)}(x_\nu) - F^{(p+1)}(X_\nu)\right| = \left|F^{(p+2)}(t)(x_\nu - X_\nu)\right|.$$

For $|x_\nu - X_\nu|$, recall that $|f'(x_\nu) - F'(X_\nu)| < \varepsilon |F'(X_\nu)| \iff |\nu - yx_\nu^{-s}| < \varepsilon y x_\nu^{-s}$. We have $|x_\nu - X_\nu| \ll \varepsilon\eta\nu^{-\sigma}$. Therefore,

$$\begin{aligned}\left|F^{(p+1)}(x_\nu) - F^{(p+1)}(X_\nu)\right| &= \left|F^{(p+2)}(t)(x_\nu - X_\nu)\right| && (9)\\ &\ll \varepsilon y N^{-s-p-1}\eta\nu^{-\sigma} \ll \varepsilon y N^{-s-p}. && (10)\end{aligned}$$

Together with the known condition $\left|f^{(p+1)}(x_\nu) - F^{(p+1)}(x_\nu)\right| < \varepsilon \left|F^{(p+1)}(x_\nu)\right| \ll \varepsilon y N^{-s-p}$, we derive that

$$\left|f^{(p+1)}(x_\nu) - F^{(p+1)}(X_\nu)\right| \ll \varepsilon y N^{-s-p}.$$

Now consider the difference of functional,

$$\begin{aligned}\left|\phi^{(p+1)}(\nu) - \Phi^{(p+1)}(\nu)\right| &= \left|\underbrace{\frac{1}{(f''(x_\nu))^{2p-1}}}_{A}\underbrace{\sum_{\mathbf{u}} w_{\mathbf{u}} \prod_{i=1}^{p-1} f^{(u_i+1)}(x_\nu)}_{S_1} - \underbrace{\frac{1}{(F''(X_\nu))^{2p-1}}}_{B}\underbrace{\sum_{\mathbf{u}} w_{\mathbf{u}} \prod_{i=1}^{p-1} F^{(u_i+1)}(X_\nu)}_{S_2}\right| && (11)\\ &= (A-B)S_1 + B(S_1 - S_2) && (12)\\ &\ll \frac{\varepsilon}{(F''(X_\nu))^{2p-1}} y^{p-1} X_\nu^{-s(p-1)-\sum_i u_i} && (13)\\ &+ \frac{1}{(F''(X_\nu))^{2p-1}} \sum_{\mathbf{u}} w_{\mathbf{u}} \left(\prod_{i=1}^{p-1} f^{(u_i+1)}(x_\nu) - \prod_{i=1}^{p-1} F^{(u_i+1)}(X_\nu)\right) && (14)\\ &\ll \frac{\varepsilon}{(F''(X_\nu))^{2p-1}} y^{p-1} X_\nu^{-s(p-1)-2p+2} && (15)\end{aligned}$$

in which the

$$\begin{aligned}\prod_{i=1}^{p-1} f^{(u_i+1)}(x_\nu) - \prod_{i=1}^{p-1} F^{(u_i+1)}(X_\nu) &= \sum_{j=1}^{p-1} \prod_{i<j} f^{(u_i+1)}(x_\nu) \prod_{i>j} F^{(u_i+1)}(x_\nu)(f^{(u_i+1)}(x_\nu) - F^{(u_i+1)}(x_\nu)) && (16)\\ &\ll \sum_{j=1}^{p-1} \prod_{i} \max_{i\neq j}(f^{(u_i+1)}(x_\nu), F^{(u_i+1)}(x_\nu)) \cdot \varepsilon y N^{-s-p} && (17)\\ &\ll \varepsilon y^{p-1} X_\nu^{-s(p-1)-2p+2}. && (18)\end{aligned}$$

We convert the $F''(X_\nu)$ into its upper bound to obtain

$$\frac{\varepsilon}{(yX_\nu^{-s-1})^{2p-1}} y^{p-1} X_\nu^{-s(p-1)-2p+2} \ll \varepsilon y^{-p} X_\nu^{ps+1} \ll \varepsilon y^{-p} N^{ps+1} \asymp \varepsilon \left|\Phi^{(p+1)}(\nu)\right|,$$

because the $X_\nu \in [X_{2N}, X_N] \asymp N^{-\sigma}$. □

The Main Proof

Theorem 4.2 (Van der Corput **B-process**). *$\forall(k,l)$ as exponent pairs, we have*

$$B(k,l) := \left(l - \frac{1}{2}, k + \frac{1}{2}\right)$$

as a new exponent pair.

Proof. We would first use the final form of Lemma 3.6 to decipher the $S = \sum \mathrm{e}(f(n))$, then derive the corresponding bound from the result of the lemma we just proved. In the proof, the function F, Φ, ϕ is always the same as in the previous lemma.

From the $B(k,l)$, we know that $0 \leqslant l - \frac{1}{2} \leqslant \frac{1}{2} \leqslant k + \frac{1}{2}$; if the $l = \frac{1}{2}$, then $k = \frac{1}{2}$, and $B(k,l) = (0,1)$. We may assume $l > \frac{1}{2}$.

According to the definition of the exponent pair, it ensures that $\exists \mathbf{F}$ such that $f \in \mathbf{F}(N, P, s, y, \varepsilon)$. In the form of inequality, $\left|f^{(p+1)}(x) - (-1)^p (s)_p y x^{-s-p}\right| < \varepsilon(s)_p y x^{-s-p} \iff f^{(p+1)}(x) \asymp LN^{-p}$, which satisfies the requirement of Lemma 3.6 about the restriction for $2, 3, 4$ order of derivatives if we set $F = LN$.

We plug the f in the equation:

$$S = \underbrace{\sum_{\alpha \leqslant \nu \leqslant \beta} \frac{\mathrm{e}(-\phi(\nu)) - 1/8}{|f''(x_\nu)|^{\frac{1}{2}}}}_{\Sigma_1} + O(\log(L+2) + L^{-\frac{1}{2}} N^{\frac{1}{2}}). \tag{19}$$

Then we transform the Σ_1 into integrals

$$\Sigma_1 = \mathrm{e}\left(-\frac{1}{8}\right) \int_\alpha^\beta \frac{\mathrm{e}(-\phi(\nu))}{|f''(x_\nu)|^{\frac{1}{2}}} d[\nu] \asymp \int_\alpha^\beta \frac{1}{|f''(x_w)|^{\frac{1}{2}}} d \underbrace{\left(\overline{\sum_{\alpha \leqslant \nu < w} \mathrm{e}(\phi(\nu))}\right)}_{T(w)}. \tag{20}$$

From the exponent pair (k,l), we obtain

$$T(w) \ll (\eta J^{-\sigma})^k J^l + \eta^{-1} J^\sigma \ll N^k J^l + N^{-1}.$$

Therefore,

$$\begin{aligned}
\int_\alpha^\beta \frac{1}{|f''(x_w)|^{\frac{1}{2}}} d\overline{T(w)} &= \left.\frac{\overline{T(w)}}{|f''(x_w)|^{\frac{1}{2}}}\right|_\alpha^\beta - \int_\alpha^\beta \overline{T(w)}\, d\left(|f''(x_w)|^{-\frac{1}{2}}\right) && (21)\\
&\ll_{(1)} (N^k L^l + N^{-1}) \left((LN^{-1})^{-\frac{1}{2}} - \int_\alpha^\beta d\left(|f''(x_w)|^{-\frac{1}{2}}\right)\right) && (22)\\
&\asymp (N^k L^l + N^{-1})(LN^{-1})^{-\frac{1}{2}} && (23)\\
&\asymp L^{l-\frac{1}{2}} N^{k+\frac{1}{2}} + L^{-\frac{1}{2}} N^{-\frac{1}{2}}. && (24)
\end{aligned}$$

For (1), the $\nu \asymp J([\alpha,\beta]) \asymp f'(N) \asymp L$. Combining with the error term,

$$S \ll L^{l-\frac{1}{2}} N^{k+\frac{1}{2}} + \log(L+2) + L^{-\frac{1}{2}} N^{\frac{1}{2}},$$

in which $L^{l-\frac{1}{2}} N^{k+\frac{1}{2}}$ dominates. $\square$

5 Equidistribution of $\log n!$ mod 1

Theorem 5.1. $\forall b > 1, \{\log_b n!\}_{n=0}^{\infty}$ *is equidistributed* mod 1

Proof. We use Van der Corput's inequality and deal with the difference function. We first apply the Weyl criterion:

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathrm{e}(\alpha \log_b n!) = 0, \forall \alpha \in \mathbb{Z},$$

and we define $S(N) := \sum_{n=1}^{N} \mathrm{e}(\alpha \log_b n!))$, $f(n) := \alpha \log_b n!$

Van der Corput's inequality

$$|S(N)|^2 \ll \frac{N}{H} + \frac{N}{H} \sum_{h \leqslant H} |S_1(h)|$$

in which

$$S_1(h) = \sum_{n=1}^{N-h} \mathrm{e}(\alpha \log_b (n+h)! - \alpha \log_b n!) = \sum_{n=1}^{N-h} \mathrm{e}\left(\frac{\alpha}{\log b} \sum_{k=1}^{h} \log(n+k) \right).$$

Then, we consider the function $g(n;h) = a \sum_{k=1}^{h} \log(n+k)$, $a = \frac{\alpha}{\log b}$.

Taylor approximation and dyadic estimation

By Taylor's formula,

$$\log(n+k) = \log n + \frac{k}{n} - \frac{k^2}{2n^2} + \frac{k^3}{3n^3} + O(\frac{k^4}{n^4}).$$

Thus, we obtain

$$g(n;h) = a\left(h \log n + \frac{h(h+1)}{2n} - \frac{h(h+1)(2h+1)}{12n^2} + \frac{h^2(h+1)^2}{12n^3} + O(h^5/n^4) \right) = ah \log n + \frac{ah(h+1)}{2n} + R(n;h),$$

and $R(n;h) \ll h^3 n^{-2}$. If $h^3 n^{-2} \ll 1$, the effects from the residue can be ignored.

We define the set

$$I = \left\{ [X, 2X) : X = 2^k, 2X \leqslant N-h \right\}; \bigcup_{s \in I} s = [1, N-h], \forall s_1, s_2 \in I, s_1 \cap s_2 = \varnothing.$$

We ascertain that there are $\ll \log(N-h) - \log h$ intervals such that $R(n;h) \ll h^3 n^{-2} \ll 1$, i.e. $X \gg h^{\frac{3}{2}}$ for $[X, 2X] \in I$, and we mark these intervals as I_1; otherwise I_2.

For the intervals that the residue $R(n;h) \ll 1$, we reduce the $S_1(h)$ to

$$\sum_{n \in [X,2X] \in I_1} \mathrm{e}\left(ah \log n + \frac{ah(h+1)}{2n} \right);\ g^{(q)}(n;h) = \frac{(-1)^{q+1} ah}{n^q} + \frac{(-1)^q ah(h+1)}{n^{q+1}}.$$

$$\Longrightarrow \sum_{n \in [X,2X] \in I_1} n^{it} \cdot \mathrm{e}\left(\frac{ah(h+1)}{2n} \right), t = 2\pi ah.$$

We use Abel summation with $\sigma(u) = \sum_{n \in [X,u]} n^{it}$:

$$\sum_{n \in [X,2X] \in I_1} n^{it} \cdot \mathrm{e}\left(\frac{ah(h+1)}{2n} \right) = \sigma(2X) e^{\frac{ah(h+1)}{4X}} - \int_X^{2X} \sigma(t) \frac{d}{dt}(e^{\frac{ah(h+1)}{4t}}) dt.$$

Thus

$$\left| \sum_{n \in [X,2X] \in I_1} n^{it} \cdot \mathrm{e}\left(\frac{ah(h+1)}{2n} \right) \right| \leqslant \left(1 + \int_X^{2X} \left| \frac{d}{dt}(e^{\frac{ah(h+1)}{4t}}) \right| dt \right) \max_{n \in [X,2X] \in I_1} |\sigma(n)|,$$

which we are familiar with.

$$\sum_{n\in[X,2X]\in I_1} \mathrm{e}\left(ah\log n + \frac{ah(h+1)}{2n}\right) \ll \frac{X+ah^2}{1+|t|}.$$

For intervals that $R(n;h)$ cannot be ignored, we use the derivatives of $g(n;h)$:

$$g^{(q)}(n;h) = (-1)^{q+1}a\sum_{k\leqslant h}\frac{1}{(n+k)^q} \asymp \frac{ah}{X^q}, n\in[X,2X]\in I_2.$$

We can use the Kusmin-Landau to derive

$$\sum_{n\in[X,2X]\in I_2} \mathrm{e}(g(n;h)) \ll \frac{X}{|a|h}.$$

Reviewing S(N)

$$S_1(h) \ll \sum_{[X,2X]\in I_1}\frac{X+ah^2}{1+|t|} + \sum_{[X,2X]\in I_2}\frac{X}{|a|h} \ll \frac{N}{1+|t|} + \frac{ah^2(\log(N-h)-\log h)}{1+|t|} + \frac{h^{\frac{1}{2}}}{|a|}.$$

The bound is already accurate enough to reveal the $S(N)$ if we set $H = N^{1/3}$. Since the entire proof has nothing to do with α, b, we have proved the theorem. □

6 $\{nP(n)\}$ mod 1's equidistribution and Polynomial vectors mod$\mathbb{Z}^k$

Theorem 6.1. *$\forall P(x)\in\mathbb{R}[x]$, $P(x)$ have at least one irrational coefficient, and $\{nP(n)\}$ mod 1 is equidistributed. Give necessary and sufficient conditions on polynomials $P_1, P_2, P_3, ..., P_k \in \mathbb{R}[x]$ for the sequence of vectors*

Proof. We still apply Weyl's criterion:

$$\lim_{N\to\infty}\frac{1}{N}\sum_{n\leqslant N}\mathrm{e}(knP(n)) = 0, \forall k\in\mathbb{Z}$$

for which polynomial $P(n)$ with an irrational coefficient. We assume that $\deg P = d$. We separate the interval $[1, N]$ into dyadic:

$$I = \left\{[X,2X) : X = 2^k, 2X \leqslant N\right\}; \bigcup_{s\in I} s = [1,N], \forall s_1, s_2 \in I, s_1\cap s_2 = \varnothing.$$

Thus, the original sum $S(N)$ is turned into

$$\sum_{n\in[X,2X]\in I}\mathrm{e}(knP(n))\cdot\log N \asymp S(N).$$

Reduction by difference

We iterate the Van der Corput inequality (Generalized Van der Corput) to fit the attribute that $\Delta_{d+1}(nP(n)) = a_p(d+1)!$ is a constant in which the a_p is the coefficient of the term with the highest order, but we would use the linear function of $\Delta_d(nP(n)) = an+b$. Giving $q = d$ and $Q = 2^q$,

$$|S_X|^Q \leqslant 8^{Q-1}\left(\frac{X^Q}{H} + \frac{X^{Q-1}}{H^{2-\frac{2}{Q}}}\sum_{\mathbf{h}}|S_q(\mathbf{h})|\right)$$

What's left is the detail of the multiple sum $\sum_{\mathbf{h}}|S_q(\mathbf{h})|$.

$$\mathbf{h} = (h_1, h_2, h_3, ..., h_q), 1\leqslant h_i \leqslant H^{2^{i-q}}; \prod_{i=1}^{q} h_i \leqslant H^{2-2/Q}; S_q(\mathbf{h}) = \sum_{n\in[X,2X-\sum h_i]} e(f_q(n;\mathbf{h})).$$

Since we can derive that for a polynomial P of degree d, $\Delta_{h_1h_2\ldots h_d}P(n) = a_p d!\prod_{i\leqslant d} h_i$,

$$
\begin{aligned}
f_q(n;\mathbf{h}) &= \Delta_{h_1h_2h_3\ldots h_q} knP(n) = kA(\mathbf{h})n + C(\mathbf{h}),\\
A(\mathbf{h})(n + h_{q+1} - n) &= a_p(q+1)!\prod_{i=1}^{q+1} h_i\\
\Longrightarrow kA(\mathbf{h})n + C(\mathbf{h}) &= ka_p(q+1)!\prod_{i=1}^{q} h_i \cdot n + C(\mathbf{h}).
\end{aligned}
$$

It is merely a linear function of n. Thus, our inequality is reduced to a rather simple form:

$$
|S_X| \ll_Q \left(\frac{X^Q}{H} + \frac{X^{Q-1}}{H^{2-\frac{2}{Q}}}\sum_{\mathbf{h}}\left|\sum_{n\in[X,2X-\sum h_i]} e\left(ka_p(q+1)!\prod_{i=1}^{q} h_i\cdot n\right)\right|\right)^{1/Q}
$$

Estimation of Linear n

By the classical arguments that $\sum_{n\in[N,2N]} e(\alpha n) = O(\|\alpha\|^{-1})$, we conclude that the sum

$$
\sum_{n\in[X,2X-\sum h_i]} e\left(A(\mathbf{h})n\right) \ll \min\left(\frac{1}{\|ka_p(q+1)!\prod_{i=1}^{q} h_i\|}, X\right).
$$

Then, we introduce some new notation: $\Pi := \prod_{i=1}^{q} h_i$, $\theta = ka_p(q+1)! \in \mathbb{R}\backslash\mathbb{Q}$. Considering the outer sum,

$$
\sum_{h_1,h_2,\ldots,h_q} \min(\|\theta\Pi\|^{-1}, X),
$$

and we remove the h variable by using Π:

$$
\sigma := \sum_{\Pi\leqslant H^{2-2/Q}} r(\Pi)\min(\|\theta\Pi\|^{-1}, X).
$$

We prove the following lemma.

Lemma 6.2.

$$
\sum_{1\leqslant\Pi\leqslant U} \min(\|\theta\Pi\|^{-1}, X) \ll (X+U)\log 2U
$$

Proof. Applying dyadic intervals,

$$
\begin{aligned}
\sum_{1\leqslant\Pi\leqslant U} \min(\|\theta\Pi\|^{-1}, X) &= \sum_{1\leqslant\Pi\leqslant U} N\cdot \mathbf{1}_{\|\theta\Pi\|\leqslant X^{-1}} + (\|\theta\Pi\|)^{-1}\cdot \mathbf{1}_{\|\theta\Pi\|\geqslant X^{-1}}\\
&\leqslant \sum_{1\leqslant\Pi\leqslant U}\left(N\cdot \mathbf{1}_{\|\theta\Pi\|\leqslant X^{-1}} + \sum_{k=1}^{[\log_2 X]+1} 2^k\cdot \mathbf{1}_{\|\theta\Pi\|\in[2^{-k},2^{-k+1}]}\right)\\
&\ll X(\frac{U}{X}+1) + \sum_{k\leqslant \log_2 X} 2^k(\frac{U}{2^k}+1) \ll (X+U)\log 2U
\end{aligned}
$$

□

Thus, we are able to deal with the sum

$$
\begin{aligned}
\sigma &\leqslant \sum_{\Pi\leqslant H^{2-2/Q}} r(\Pi)\min(\|\theta\Pi\|^{-1}, X) = \sum_{\Pi\leqslant H^{2-2/Q}} r(\Pi)\left(N\cdot \mathbf{1}_{\|\theta\Pi\|\leqslant X^{-1}} + (\|\theta\Pi\|)^{-1}\cdot \mathbf{1}_{\|\theta\Pi\|\geqslant X^{-1}}\right)\\
&\ll_{(1)} H^{(2-2/Q)\varepsilon}(X + H^{2-2/Q})\log 2H^{2-2/Q}, \forall\varepsilon > 0.
\end{aligned}
$$

In (1), we deduce that the $r(\Pi) := \#\{\mathbf{h} : \prod h_i = \Pi,\ 1\leqslant h_i\leqslant H^{2^{i-q}}\}$ is bounded by $\ll_\varepsilon \Pi^\varepsilon \leqslant H^{(2-2/Q)\varepsilon}$, which is actually loose for each h_i subjected to tight conditions ($H^{2^{i-q}}$), and the following bound is obtained by Lemma 3.2.

6.1 Reviewing S(N)

We take the result back to the $|S_X|^Q$:

$$|S_X| \ll_Q \left(\frac{X^Q}{H} + \frac{X^{Q-1}}{H^{2-\frac{2}{Q}}} \sigma \right)^{1/Q} \ll \left(\frac{X^Q}{H} + \frac{X^{Q-1}}{H^{2-\frac{2}{Q}}} H^{(2-2/Q)\varepsilon} (X + H^{2-2/Q}) \log 2H^{2-2/Q} \right)^{1/Q}.$$

We found that if we set $H = X^\gamma$, the bound can be reduced to $X^{Q-\kappa(\gamma)}$. Therefore, we conclude that the $S(N) = o(N) \iff a_p \in \mathbb{R} \backslash \mathbb{Q}$. $\square$

7 Exercises from *254A, Notes 5: Bounding exponential sums and the zeta function*

Prerequisite Theorems:

Theorem 7.1. *For $s=\sigma+it,\ 0<\varepsilon\leqslant\sigma\ll 1,\ |t|\gg 1,$*

$$\zeta(s)\ll_{\varepsilon}\log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{1-\sigma}\left|\frac{1}{N}\sum_{N\leqslant n<N+M}\mathrm{e}\Big(-\frac{t}{2\pi}\log n\Big)\right| \tag{25}$$

from the dyadic method and smooth cutoff.

Theorem 7.2. $2\leqslant N\ll T, |I|<N,\ \left|f^{(j)}(x)\right|=\exp(O(j^2))\frac{T}{N^j}$. *Then,*

$$i)\ (\textit{Van der Corput's estimate})\ \frac{1}{N}\sum_{n\in I}\mathrm{e}(f(n))\ll\left(\frac{T}{N^k}\right)^{\frac{1}{2^k-2}}\log^{\frac{1}{2}}(T+2),\ \ \forall k\geqslant 2 \tag{26}$$

$$ii)\ (\textit{Vinogradov's estimate})\ \frac{1}{N}\sum_{n\in I}\mathrm{e}(f(n))\ll N^{-\frac{c}{k^2}},\ \ \forall T\leqslant N^k, c\textit{ is absolute constant.} \tag{27}$$

The Van der Corput's estimate stated in the Theorem can be properly depicted by terminologies from exponent pairs, which is the A-process (Theorem 5.3). The result in the theorem is equivalent to

$$(\kappa_k,\lambda_k)=\left(\frac{1}{2^k-2},1-\frac{k}{2^k-2}\right)\text{ is an exponent pair,}$$

but $(\kappa_k,\lambda_k)=A^k(0,1)$.

7.1 Exercise 3 (Subconvexity bound)

Theorem 7.3. *(i) Show that $\zeta\left(\frac{1}{2}+it\right)\ll(1+|t|)^{1/6}\log^{O(1)}(2+|t|)$ for all $t\in\mathbb{R}$. (Hint: use the $k=3$ case of the Van der Corput estimate)*
(ii) For any $0<\sigma<1$, show that $\zeta(\sigma+it)\ll(1+|t|)^{\max(\frac{1-\sigma}{3},\frac{1}{2}-\frac{2\sigma}{3})+o(1)}$ as $|t|\to\infty$ (the decay rate in the $o(1)$ is allowed to depend on σ).

Proof. For (i),

$$\zeta\left(\frac{1}{2}+it\right)\ll_{\varepsilon}\log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{\frac{1}{2}}\left|\frac{1}{N}\sum_{N\leqslant n<N+M}e(-\frac{t}{2\pi}\log n)\right|,$$

and for function $f(n)=-\frac{t}{2\pi}\log n,\ \left|f^{(j)}(x)\right|\asymp\frac{|t|}{N^j}$. Then, we apply van der Corput's estimate when $k=3$.

$$\begin{aligned}\zeta\left(\frac{1}{2}+it\right)&\ll\log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{\frac{1}{2}}\left(\frac{|t|^{\frac{1}{6}}}{N^{\frac{1}{2}}}\log^{\frac{1}{2}}(|t|+2)\right)\\&\asymp|t|^{\frac{1}{6}}\log^{\frac{3}{2}}(|t|+2)\\&\ll(1+|t|)^{\frac{1}{6}}\log^{O(1)}(|t|+2)\end{aligned}$$

For (ii), we have to separate the interval $[N,2N)$. When van der Corput's estimate offers a result that is worse than the trivial bound, we replace it with trivial bound.

$$\left(\frac{T}{N^k}\right)^{\frac{1}{2^k-2}}\log^{\frac{1}{2}}(T+2)\gg 1\Longleftrightarrow N\ll R_k(T):=T^{\frac{1}{k}}\log^{\frac{1}{2k(2^k-2)}}(T+2).$$

Thus, we apply the $B(T)\asymp T^{\frac{1}{k}+o(1)}$ to the ζ function's exponential sum $S_M=\frac{1}{N}\sum_{N\leqslant n<N+M}e(-\frac{t}{2\pi}\log n)$:

$$\zeta(s)\ll_{\varepsilon}\log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{1-\sigma}\left|S_M\right| \tag{28}$$

$$\ll\log(|t|+2)\min_{1<N_1\ll R_k(|t|)\ll N_2\ll|t|}\left(N_1^{1-\sigma},|t|^{\frac{1}{2^k-2}}N_2^{1-\sigma-\frac{k}{2^k-2}}\log^{\frac{1}{2}}(|t|+2)\right) \tag{29}$$

$$\ll\left(R_k^{1-\sigma}(|t|)+|t|^{\frac{1}{2^k-2}}\log^{\frac{1}{2}}(|t|+2)\max_{R_k(|t|)\ll N\ll|t|}N^{1-\sigma-\frac{k}{2^k-2}}\right)\log(|t|+2) \tag{30}$$

Now, what's remains to be debunked is the $N^{1-\sigma-\frac{k}{2^k-2}}$. If $1-\sigma-\frac{k}{2^k-2}<0$, then the (30) is equivalent to $|t|^{\frac{1-\sigma}{k}+o(1)}$; if $1-\sigma-\frac{k}{2^k-2}>0$, then the (30) is $|t|^{\max(1-\sigma-\frac{k-1}{2^k-2},\frac{1-\sigma}{k})+o(1)}$. To sum up, we have derived $|t|^{\max(1-\sigma-\frac{k-1}{2^k-2},\frac{1-\sigma}{k})+o(1)}$ □

7.2 Exercise 4

Theorem 7.4. *Let t be such that $|t|\geqslant 100$, and let $\sigma\geqslant 1/2$.*
(i) (Littlewood bound) Use the van der Corput estimate to show that

$$\zeta(\sigma+it)\ll \log^{O(1)}|t|\,,\ \textit{whenever}\ \sigma\geqslant 1-O\bigg(\frac{(\log\log|t|)^2}{\log|t|}\bigg).$$

(ii) (Vinogradov-Korobov bound) Use the Vinogradov estimate to show that

$$\zeta(\sigma+it)\ll \log^{O(1)}|t|\,,\ \textit{whenever}\ \sigma\geqslant 1-O\Bigg(\frac{(\log\log|t|)^{2/3}}{\log^{2/3}|t|}\Bigg).$$

Proof. For the Littlewood bound, we derive

$$\zeta(s)\ll_\varepsilon \log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{O(\delta)}\,|S_M|\,,\quad \delta:=\frac{(\log\log|t|)^2}{\log|t|}$$

due to $1-\sigma\leqslant O(\frac{(\log\log|t|)^2}{\log|t|})$, in which the S_M is the same as in exercise 3. We incorporate the train of thought in exercise 3 to estimate $\zeta(s)$, with $R_k(T)\asymp T^{\frac{1}{k}}$.

$$\begin{aligned}\zeta(s)&\ll \log(|t|+2)\min_{1<N_1\ll R_k(|t|)\ll N_2\ll|t|}\left(N_1^{O(\delta)},|t|^{\frac{1}{2^k-2}}\,N_2^{O(\delta)-\frac{k}{2^k-2}}\log^{\frac12}(|t|+2)\right) &(31)\\ &\ll \log^{O(1)}|t|\min_{1<N_1\ll R_k(|t|)\ll N_2\ll|t|}\left(N_1^{O(\delta)},|t|^{\frac{1}{2^k-2}}\,N_2^{O(\delta)-\frac{k}{2^k-2}}\right) &(32)\\ &\ll \log^{O(1)}|t|\left(R_k(|t|)^{O(\delta)}\,|t|^{\frac{1}{2^k-2}}\,R_k(|t|)^{O(\delta)-\frac{k}{2^k-2}}\right)^{\frac12} &(33)\end{aligned}$$

We arrive at $R_k(|t|)^{O(\delta)}\,|t|^{\frac{1}{2^k-2}}\,R_k(|t|)^{O(\delta)-\frac{k}{2^k-2}}$ because $\min(a,b)\leqslant(ab)^{\frac12}$, and since the functions minimizing are monotonically increasing $f(N_1)=N_1^{O(\delta)}$, and monotonically decreasing $g(N_2)=|t|^{\frac{1}{2^k-2}}\,N_2^{O(\delta)-\frac{k}{2^k-2}}$, the maximum appears at the boundary $R_k(|t|)$ for the entire minimum, which means $N_1\asymp R_k(|t|)\asymp N_2$. Set $k=[c\log\log|t|]$ for some absolute constant c,

$$\begin{aligned}(33)\iff&\log^{O(1)}|t|\left(|t|^{\frac{1}{2^k-2}}\,|t|^{O(\delta/k)-\frac{1}{2^k-2}}\right)^{\frac12}\\ \asymp&\left(|t|^{O(\delta/k)}\right)^{\frac12}\log^{O(1)}|t|\\ \asymp&|t|^{\frac{\log\log|t|}{\log|t|}}\log^{O(1)}|t|\asymp\log^{O(1)}|t|\,.\end{aligned}$$

For Vinogradov-Korobov bound, we apply Vinogradov's estimate, which is applicable only if $T\ll N^k$. We define some quantities first:

$$S_M:=\frac{1}{N}\sum_{N\leqslant n<N+M}\mathrm{e}\bigg(-\frac{t}{2\pi}\log n\bigg)\,,\quad \delta:=\frac{(\log\log|t|)^{2/3}}{\log^{2/3}|t|},\quad k=[\log\log|t|].$$

By classical arguments,

$$\begin{aligned}\zeta(s)&\ll_\varepsilon \log(|t|+2)\sup_{1<M\leqslant N\ll|t|}N^{1-\sigma}\,|S_M|\\ &\ll \log(|t|+2)\min_{1<N_1\ll|t|^{1/k}\ll N_2\ll|t|}\left(N_1^{O(\delta)},N_2^{O(\delta)-\frac{c}{k^2}}\right),\end{aligned}$$

for some absolute constant c. The Vinogradov-estimate can deal with the $N_2^k \gg |t|$ just right. Since $f(N) = N^{O(\delta)}$ monotonically increases and $g(N) = N^{O(\delta)-\frac{c}{k^2}}$ monotonically decreases, the $\sup\min_{N_1 \ll |t|^{1/k} \ll N_2} \iff N_1 \asymp N_2 \asymp |t|^{1/k}$.

$$\begin{aligned}\zeta(s) &\ll \log(|t|+2)\min\left(|t|^{O(\delta/k)}, |t|^{O(\delta/k)-\frac{c}{k^3}}\right) \\ &\ll |t|^{O(\delta/k)-\frac{c}{k^3}}\log(|t|+2) \ll \log^{O(1)}|t|.\end{aligned}$$

□

7.3 Remarks on the choice of k and technical points in the Littlewood bound

Remark 7.5 (Choice of the parameter k in van der Corput's method)**.** *In the proof of the Littlewood bound for $\zeta(\sigma+it)$, the parameter k in the k-th derivative test of van der Corput's method must be chosen with care. Although the bound obtained from the oscillatory integral formally improves as k increases, the implicit constant in the van der Corput estimate grows rapidly with k.*

We reconstruct the proof procedure of Weyl-differencing, the fundamental principal of A-process.

1. $$|S|^2 = \sum_h \sum_n e(f(n+h)-f(n)) \Rightarrow |S|^2 \leqslant \sum_{|h|\leqslant H}\left|\sum_n e(\Delta_h f(n))\right| + \textit{(error)},$$

   losing factor H. Cauchy-Schwarz,

   $$|S|^4 \ll H\sum_h \left|\sum_n e(\Delta_h f(n))\right|^2,$$

   add one H to the constant C_k.

2. *Iterate k times, we derive*

   $$|S|^{2^k} \ll \sum_{h_1,h_2,\ldots,h_k}\left|\sum_n e(\Delta_{h_1}\Delta_{h_2}\cdots\Delta_{h_k} f(n))\right|.$$

   Considering the difference function,

   $$\Delta_{h_1}\Delta_{h_2}\cdots\Delta_{h_k} f(n) = \sum_{J\subseteq\{1,\ldots,k\}} (-1)^{k-|J|} f\Big(n + \sum_{j\in J} h_j\Big).$$

   The k-fold difference involves 2^k shifted values of f, reflecting an exponential combinatorial complexity.

3. *We can also analyze the Taylor expansion of difference since we have $f(x+h) = f(x) + f'(x)h + o(h)$:*

   $$\Delta_{h_1}\Delta_{h_2}\cdots\Delta_{h_k} f(n) = \frac{f^{(k)}(n)}{k!} h_1 h_2 h_3 \cdots h_k + O\left(\sum_i |h_i|^{k+1}\right).$$

   We must at least multiply $k!$ to obtain effective oscillation of function f.

To sum up, we have a c^k for Cauchy-Schwarz, a $\prod_i H_i \asymp c^k$, and a $k!$ from differencing function. For a constant c,

$$C_k \gtrsim c^k k! \asymp c^k k^{k+\frac{1}{2}} e^{-k} \asymp (Ck)^k. \tag{34}$$

Remark 7.6 (The critical scale $R_k(t)$)**.** *We know $R_k(t) \asymp |t|^{1/k}$. In the estimation of $\zeta(s)$ via short exponential sums, one naturally encounters expressions of the form*

$$\min_{1<N_1\ll R_k(t)\ll N_2 \ll |t|}\left(N_1^{O(\delta)}, |t|^{\frac{1}{2^k-2}} N_2^{O(\delta)-\frac{k}{2^k-2}}\right),$$

The first term is monotone increasing in N_1, while the second term is monotone decreasing in N_2. Hence the worst (largest) value of the minimum occurs at the boundary

$$\sup \min_{N_1 \ll R_k(|t|) \ll N_2} (f(N_1), g(N_2)) \iff N_1 \asymp N_2 \asymp R_k(t).$$

This explains why the scale $R_k(t)$ is intrinsic to the argument and cannot be avoided.

Remark 7.7 (Common pitfalls). *In applying van der Corput's method to the Littlewood bound, the following points are essential:*

- *One must not take k to grow faster than $\log\log|t|$, since the implicit constant C_k grows at least as fast as $(Ak)^k$.*
- *The appearance of negative powers of $|t|$ in intermediate expressions should not be overinterpreted; at the critical scale $R_k(t)$, these powers cancel exactly.*
- *The final bound is of the form*
$$\zeta(\sigma+it) \ll |t|^{O(\delta/k)} \log^{O(1)} |t| \,,$$
and since $\delta/k = o(1)$ for $k \asymp \log\log|t|$, this yields the Littlewood bound
$$\zeta(\sigma+it) \ll \log^{O(1)} |t| \,.$$

7.4 Exercise 5 (Vinogradov-Korobov in arithmetic progressions)

Theorem 7.8. *Let χ be a non-principal character of modulus q.*

(i) (Vinogradov-Korobov bound) Use the Vinogradov estimate to show that
$$L(\sigma+it,\chi) \ll \log^{O(1)}(q\,|t|)$$
whenever $|t| \geqslant 100$ and
$$\sigma \geqslant 1 - O\left(\min\left(\frac{\log\log(q\,|t|)}{\log q}, \frac{(\log\log(q|t|))^{2/3}}{\log^{2/3}|t|}\right)\right).$$
(Hint: use the Vinogradov estimate and a change of variables to control $\sum_{n\in I: n\equiv a \pmod q} \exp(-it\log n)$ for various intervals I of length at most N and residue classes $a \pmod q$, in the regime $N \geqslant q^2$ (say). For $N < q^2$, do not try to capture any cancellation and just use the triangle inequality instead.)

(ii) Obtain a zero-free region
$$\left\{\sigma+it : \sigma > 1 - c\min\left(\frac{1}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \frac{1}{\log q}\right),\ |t| \geqslant 200\right\}$$
for $L(s,\chi)$, for some (effective) absolute constant $c>0$.

(iii) Obtain the prime number theorem in arithmetic progressions with an error term
$$\sum_{\substack{n\leqslant x\\ n\equiv a \pmod q}} \Lambda(n) = \frac{x}{\phi(q)} + O\left(x\exp\left(-c_A\frac{\log^{3/5}x}{(\log\log x)^{1/5}}\right)\right),$$
whenever $x>100$, $q\leqslant \log^A x$, $a \pmod q$ is primitive, and $c_A>0$ depends (ineffectively) on A.

Proof. The most crucial obstacle is propagating the result of ζ to $L(s,\chi)$.

7.4.1 Vinogradov-Korobov bound of Dirichlet $L(s,\chi)$

For (i), we first analyze the $L(s,\chi), s=\sigma+it$. We will try to prove a lemma that is quite similar to Theorem 7.1. By some transformation,
$$L(s,\chi) = \sum_n \frac{\chi(n)}{n^s} = \sum_{\substack{a\leqslant q\\(a,q)=1}} \chi(a) \sum_{n\equiv a \pmod q} \frac{1}{n^s} \tag{35}$$
$$= \sum_{\substack{a\leqslant q\\(a,q)=1}} \chi(a) \sum_n \frac{1}{(a+qn)^s} = q^{-s} \sum_{\substack{a\leqslant q\\(a,q)=1}} \chi(a)\zeta\left(s,\frac{a}{q}\right), \tag{36}$$

we reduce the study of $L(s,\chi)$ to a linear combination of Hurwitz zeta functions. We shall use $\alpha = \frac{a}{q}$ for a given and fixed a.

We define the smooth cut-off function:

$$\eta(x) = \begin{cases} 0 & \text{for} \quad x > C \\ 1 & \text{for} \quad x < -C \\ C^\infty(\mathbb{R}) & \text{for} \quad x \in [-C, C] \end{cases}.$$

And we introduce the function

$$f(s) := \sum_n \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log x) + \int_{\mathbb{R}} \frac{(t+\alpha)^{-s}}{1-s}\eta'(\log(t+\alpha) - \log x)dt \tag{37}$$

$$= \sum_n \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log x) + \int_{\mathbb{R}} \frac{e^{(1-s)u}}{1-s}\eta'(u - \log x)du \quad (u = \log(t+\alpha)) \tag{38}$$

$$= \sum_n \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log x) + x^{1-s}\int_{\mathbb{R}} \frac{e^{(1-s)t}}{1-s}\eta'(t)dt \tag{39}$$

It would converge uniformly to $\zeta(s,\alpha)$ in the region $\{s : \Re(s) > 0; s \neq 1\}$ for trivial arguments shown in Exercise 32 of *254A, Supplement 3: The Gamma function and the functional equation.* For critical strip $\{s : \Re(s) \in [0,1]\}$, the integral $x^{1-s}\int_{\mathbb{R}} \frac{e^{(1-s)t}}{1-s}\eta'(t)dt = -x^{1-s}\int_{\mathbb{R}} e^{(1-s)t}\eta(t)dt.$

Lemma 7.9. *Similar to the $\zeta(s)$ scenario, we deduce*

$$\sum_n \frac{1}{(n+\alpha)^s}\psi(\log(n+\alpha) - \log x) - x^{1-s}\int_{\mathbb{R}} e^{(1-s)t}\psi(t)dt = O_{\psi,A}(x^{-A})$$

for s in the critical strip $\{s : 0 < \Re(s) < 1\}$. In addition, $x \gg_\psi 1 + |t|$, and $\psi(u)$ here is a smooth, compactly supported function ($\exists$ constants $A < B$, $\psi(u) \neq 0 \Leftrightarrow u \in [A,B]$; $\quad \psi \in C^\infty((A,B))$).

Proof. The integral $\int_{\mathbb{R}} e^{(1-s)t}\psi(t)dt = O_{\psi,A}(x^{-A})$ can be derived from $\int_{\mathbb{R}} \frac{1}{(t+\alpha)^s}\psi(\log(t+\alpha) - \log x)dt$, which is $x^{1-s}\int_{\mathbb{R}} \frac{e^{(1-s)t}}{1-s}\psi'(t)dt$ for critical strip. We apply Poisson Summation Formula due to $n+\alpha \asymp x \gg_\psi 1 + |t|$,

$$\sum_n \frac{1}{(n+\alpha)^s}\psi(\log(n+\alpha) - \log x) = \sum_{m\in\mathbb{Z}} H(m),$$
$$H(m) := \int_{\mathbb{R}} \frac{1}{(y+\alpha)^s}\psi(\log(y+\alpha) - \log x) \cdot e(m(y+\alpha))dy.$$

We substitute s with $\sigma + it$ and $u = \log(y+\alpha) - \log x$ with y.

$$H(m) = x^{1-s}\int_{\mathbb{R}} e^{(1-s)u}\psi(u)e(mxe^u)du$$
$$= x^{1-s}\int_{\mathbb{R}} \underbrace{e^{(1-\sigma)u}\psi(u)}_{\hat\psi(u)} \mathrm{e}\left(mx\underbrace{\left(e^u - \frac{t}{2\pi mx}u\right)}_{h_m(u)}\right)du = x^{1-s}\int_{\mathbb{R}} \hat\psi(u)\,\mathrm{e}(mxh_m(u))\,du.$$
$$H(0) = \int_{\mathbb{R}} \frac{1}{(y+\alpha)^s}\psi(\log(y+\alpha) - \log x)dt.$$

Thus, we reduce the lemma to

$$\sum_{m\in\mathbb{Z}} H(m) - H(0) \ll_{\psi,A} x^{-A}.$$

Now, the proof is completely the same as $\zeta(s)$.

$$\sum_{m\in\mathbb{Z}} H(m) - H(0) \ll_{\psi,A} x^{-A} \Longleftrightarrow \int_{\mathbb{R}} \hat\psi(u)\,\mathrm{e}(mxh_m(u))\,du \ll_{\psi,A} (|m|\,x)^{-A}.$$

Since the $\psi(u)$ confines the growth of $u \ll_C 1$, $h_m^{(k)}(u) = O_{\psi,k}(1)$. We conduct integration by-parts for the integral asymptotically in support of $\psi(u)$: $\mathrm{e}(mxh_m(u))\, du = \frac{1}{2\pi i m x h'_m(u)} d(\mathrm{e}(mxh_m(u)))$.

$$\int_{\mathbb{R}} \hat{\psi}(u)\,\mathrm{e}(mxh_m(u))\, du \asymp_\psi \frac{1}{mx}\int_{\mathbb{R}} \hat{\psi}(u) d(\mathrm{e}(mxh_m(u))) \asymp_\psi \frac{1}{mx}\left(1 - \int_{\mathbb{R}} \hat{\psi}'(u)\,\mathrm{e}(mxh_m(u))\, du\right) \asymp_\psi \cdots \ll_{\psi,A} x^{-A}.$$

$\square$

We now incorporate Lemma 7.9. to prove the following result

Theorem 7.10. *Hurwitz $\zeta(s,\alpha)$ has similar properties to Riemann $\zeta(s)$.*

$$\zeta(s,\alpha) = \sum_n \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log x) - x^{1-s}\int_{\mathbb{R}} e^{(1-s)u}\eta(u)du + O_{\eta,A,\sigma}(x^{-A}) \tag{40}$$

Proof. Assume that ψ is a smooth, compactly supported function that equals η when $u \in [-C, C]$, we obtain

$$\begin{aligned} f(s) &= \left(\sum_{n<e^C x} \frac{1}{(n+\alpha)^s} - x^{1-s}\int_{-\infty}^{-C} e^{(1-s)u} du\right) + \left(\sum_n \frac{1}{(n+\alpha)^s}\psi(\log(n+\alpha) - \log x) - x^{1-s}\int_{\mathbb{R}} e^{(1-s)t}\psi(t)dt\right) \\ &= \left(\sum_{n<e^C x} \frac{1}{(n+\alpha)^s} - \frac{x^{1-s}e^{-(1-s)C}}{1-s}\right) + O_{\eta,A,\sigma}(x^{-A}) \\ &\sim \zeta(s,\alpha) + O_{\eta,A,\sigma}(x^{-A}), \quad (x\to\infty). \end{aligned}$$

$\square$

Choosing $x \asymp |t| + 1$, the main term $x^{1-s}\int e^{(1-s)u}\eta(u)\, du$ is $O_{\eta,\sigma}(1)$. Hence, we obtain the truncated representation $\zeta(s,\alpha) = \sum_n (n+\alpha)^{-s}\eta(\log(n+\alpha) - \log x) + O_{\eta,\sigma}(1)$.

Theorem 7.11. *We deduce the following bound of Hurwitz $\zeta(s,\alpha)$ for $s = \sigma + it$, $0 < \varepsilon \leqslant \sigma \ll 1$, and $|t| \gg 1$.*

$$\zeta(s,\alpha) \ll \log(|t|+2) \sup_{1\leqslant M\leqslant N\ll|t|} N^{-\sigma}\left|\sum_{N\leqslant n<N+M} \mathrm{e}\left(-\frac{t}{2\pi}\log(n+\alpha)\right)\right|.$$

Proof. We know the result

$$\zeta(s,\alpha) = \sum_n (n+\alpha)^{-s}\eta(\log(n+\alpha) - \log(C\,|t|)) + O_{\eta,\sigma}(1)$$

for some constant C from the last theorem. Utilizing dyadic decomposition,

$$\zeta(s,\alpha) \ll 1 + \log(|t|+2)\cdot \sup_{1<N\ll|t|}\left|\sum_{N\leqslant n<2N} \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log(C\,|t|))\right| \tag{41}$$

We seek to transform the $\sum_{N\leqslant n<2N} \frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log(C\,|t|))$ into exponential sums.

$$\begin{aligned} &\frac{1}{(n+\alpha)^s}\eta(\log(n+\alpha) - \log(C\,|t|)) = N^{-\sigma}\,\mathrm{e}\left(-\frac{t}{2\pi}\log(n+\alpha)\right)F_N(n) \\ &F_N(n) := \left(\frac{N}{n+\alpha}\right)^\sigma \eta(\log(n+\alpha) - \log(C\,|t|)). \end{aligned}$$

It thus suffices to show that

$$\sum_{N\leqslant n<2N} \mathrm{e}\left(-\frac{t}{2\pi}\log(n+\alpha)\right)F_N(n) \ll \sup_{1\leqslant M\leqslant N}\left|\sum_{N\leqslant n<N+M} \mathrm{e}\left(-\frac{t}{2\pi}\log(n+\alpha)\right)\right|.$$

However, from simple Abel summation, LHS can be rewritten as:

$$F_N(2N)\sum_{N\leqslant n<2N}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\alpha)\Big)-\int_0^N\sum_{N\leqslant n<N+M}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\alpha)\Big)\cdot F_N'(M)dM$$

$$\ll\left|\sum_{N\leqslant n<2N}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\alpha)\Big)\right|+\sup_{1\leqslant M\leqslant N\ll|t|}\left|\sum_{N\leqslant n<N+M}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\alpha)\Big)\right|\int_0^N F_N'(M)dM$$

$$\asymp\sup_{1\leqslant M\leqslant N\ll|t|}\left|\sum_{N\leqslant n<N+M}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\alpha)\Big)\right|.$$

□

Now, we are ready to finish the proof by constructing a exponential sum bound for $L(s,\chi)$.

$$L(\sigma+it,\chi)\ll q^{-s}\sum_{\substack{a\leqslant q\\(a,q)=1}}\chi(a)\log(|t|+2)\sup_{1\leqslant M\leqslant N\ll|t|}N^{-\sigma}\left|\sum_{N\leqslant n<N+M}\mathrm{e}\Big(-\frac{t}{2\pi}\log(n+\frac{a}{q})\Big)\right|$$

$$\asymp q^{-\sigma}\log(|t|+2)\sum_{\substack{a\leqslant q\\(a,q)=1}}\chi(a)\sup_{1\leqslant M\leqslant N\ll|t|}N^{-\sigma}\left|\sum_{\substack{qN+a\leqslant n<q(N+M)+a\\n\equiv a\pmod q}}\mathrm{e}\Big(-\frac{t}{2\pi}\log n\Big)\right|$$

$$\ll q^{-\sigma}\log(|t|+2)\sup_{1\leqslant M\leqslant N\ll|t|}N^{-\sigma}\underbrace{\left|\sum_{n\in I_q(N,M)}\chi(n)\,\mathrm{e}\Big(-\frac{t}{2\pi}\log n\Big)\right|}_{S(I_q(N,M))},$$

in which $I_q(N,M):=\{n:\ qN\leqslant n<q(N+M)\}$, since $\chi(n)=0$ whenever $(n,q)>1$, the sum over a mod q may be recombined into a single sum over integers $n\equiv a\pmod q$.

we evaluate the sum using the Vinogradov estimate (Theorem 7.2 (ii)) with $f(x)=-\frac{t}{2\pi}\log x,\ |I_q(N,M)|\asymp qM\leqslant qN$, and we define $\delta_1=\frac{\log\log(q|t|)}{\log q},\delta_2=\frac{(\log\log(q|t|))^{2/3}}{\log^{2/3}|t|}$. Since we recognize that $f^{(k)}(n)=\exp(O(j^2))\frac{|t|}{N^k}$, we separate the range of N into $N\leqslant q^2$(trivial estimate) and $N>q^2$(Vinogradov).

$$\sup_{1\leqslant M\leqslant N\ll|t|}qMN^{-\sigma}\left(\frac{1}{qM}S(I_q(N,M))\right)$$

$$\ll q^{1+2(1-\sigma)}+\sup_{1\leqslant M\leqslant N\ll|t|}\left(\sup_{\substack{q^3<qN\leqslant|t|^{1/k}\\\text{or}\\qM<|t|^{1/k}<qN}}(qMN^{-\sigma})+\sup_{|t|^{1/k}<qM\leqslant qN}(qM)^{1-\frac{c}{k^2}}N^{-\sigma}\right)$$

$$\ll q^{1+2(1-\sigma)}+\sup_{1\leqslant M\leqslant N\ll|t|}\left(q^\sigma|t|^{\frac{1-\sigma}{k}}+q^{1-\frac{c}{k^2}}\sup_{|t|^{1/k}<qN}N^{1-\sigma-\frac{c}{k^2}}\right)$$

For the last maximum of N,

$$q^{1-\frac{c}{k^2}}\sup_{|t|^{1/k}<qN}N^{1-\sigma-\frac{c}{k^2}}=\begin{cases}q^{1-\frac{c}{k^2}}\frac{|t|^{\frac{1-\sigma}{k}-\frac{c}{k^3}}}{q^{1-\sigma-\frac{c}{k^2}}}=q^\sigma|t|^{\frac{1-\sigma}{k}-\frac{c}{k^3}} & \text{for}\quad 1-\sigma-\frac{c}{k^2}<0\quad(N=|t|^{1/k})\\ q^{1-\frac{c}{k^2}}|t|^{1-\sigma-\frac{c}{k^2}} & \text{for}\quad 1-\sigma-\frac{c}{k^2}>0\end{cases}.$$

We seek to exclude the second scenario by setting $k=\frac{(\log|t|)^{1/3}}{(\log\log|t|)^{1/3}}$. Therefore, the final bound $\log^{O(1)}(q|t|)$ is attained due to $1-\sigma\ll\min(\delta_1,\delta_2)\ll k^{-2},\ |t|^{\frac{1-\sigma}{k}}\ll|t|^{\frac{\log\log|t|}{\log|t|}}=\log^{O(1)}|t|$, and the fact that $q^{1+2(1-\sigma)}=O_q(1)$.

7.4.2 Zero-Free Region of $L(s, \chi)$

For (2), we employ the Vinogradov-Korobov bound to determine the Zero-free Region.

We deal with the $\zeta(s)$ case of the Zero-Free Region via the Vinogradov-Korobov bound (Theorem 7.4 (*ii*)) first and seek propagation.

Lemma 7.12. *When $s = \sigma + it$, $|t| \geqslant 200$, and $\sigma > 1 - \delta(t)$, the Vinogradov-Korobov bound*

$$|\zeta(s)| \ll \log^{O(1)} |t|$$

holds, in which $\delta(t) := c\frac{(\log\log|t|)^{2/3}}{\log^{2/3}|t|}$ and $c > 0$ are absolute constants. Then, for $\sigma = \Re s > 1$

$$-\frac{\zeta'}{\zeta}(s) = -\sum_{\rho:|\rho-s|\leqslant\delta(t)} \frac{1}{s-\rho} + O\left(\log^{2/3}|t|(\log\log|t|)^{1/3}\right). \tag{42}$$

The sum is over zeros ρ of $\zeta(s)$. Furthermore, the number of zeroes ρ in the above sum is $O(\log\log t)$.

Proof. We analyze $f(s) = (s-1)\zeta(s)$ instead of directly studying $\zeta(s)$, so the main function is holomorphic at $\Re s > 0$ and $\frac{f'}{f}(s) = \frac{1}{s-1} + \frac{\zeta'}{\zeta}(s)$. This lemma follows by applying Tao's *truncated log-derivative formula* to $f(s)$, together with the Vinogradov–Korobov bound.

First, we define the disk set $D(s, R) := \{z \in \mathbb{C} : |z-s| \leqslant R\}$ and utilize $D = D(1+\delta(t)+it, \delta(t))$. Furthermore, we have to construct a M that $|f(z)| \leqslant M^{O_{c_1,c_2}(1)} |f(z_0)|, z \in D, z_0 := 1+\delta(t)+it, 0 < c_1 < c_2 < 1$. From the Vinogradov-Korobov bound, we can use $M = \log^{O(1)}|t|$, and thus $\log M \asymp \log\log t$.

Normalizing $z_0 = 0$ without loss of generality, the disk is turned to $D = D(0, \delta(t))$. What's left to prove is that

$$\sum_{\rho:|\rho|\leqslant\delta(t)} \frac{1}{s-\rho} - \frac{f'}{f}(s) = O(\frac{\log M}{\delta(t)}) = O(\frac{\log\log t}{\delta(t)}).$$

Applying Tao's truncated log-derivative formula to f with radius $r = \delta(t)$, we obtain

$$\frac{f'}{f}(s) = \sum_{\rho:|\rho-s|\leqslant c_1\delta(t)} \frac{1}{s-\rho} + O\left(\frac{\log M}{\delta(t)}\right) = \sum_{\rho:|\rho-s|\leqslant c_1\delta(t)} \frac{1}{s-\rho} + O\left(\log^{2/3}|t|(\log\log|t|)^{1/3}\right).$$

Absorbing the term $\frac{1}{s-1}$ to $O(\log^{2/3}|t|(\log\log|t|)^{1/3})$ gives the desired formula for $-\zeta'(s)/\zeta(s)$. This completes the proof. ☐

Corollary 7.13. *When $s = \sigma + it$, $|t| \geqslant 200$, and $\sigma > 1 - \delta(t)$, the Vinogradov-Korobov bound*

$$|L(s,\chi)| \ll \log^{O(1)}(q\,|t|)$$

holds, in which

$$\delta(t) := c \min\left(\frac{\log\log(q\,|t|)}{\log q}, \frac{(\log\log(q|t|))^{2/3}}{\log^{2/3}|t|}\right)$$

and $c > 0$ are absolute constants. Then, for $\sigma = \Re s > 1$

$$-\frac{L'}{L}(s,\chi) = -\sum_{\rho:|\rho-s|\leqslant\delta(t)} \frac{1}{s-\rho} + O\left(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\right). \tag{43}$$

The sum is over zeros ρ of $L(s,\chi)$. Furthermore, the number of zeroes ρ in the above sum is $O(\log\log(q|t|))$.

Proof. We follow almost identical procedures as in ζ case; nonetheless, $L(s,\chi)$ is more generalized than $\zeta(s)$. We noticed that the $L(s,\chi)$ is holomorphic and pole-free in $\Re s > 0$ since χ is non-principal. After we normalize the center of the disk set to $D(0,\delta(t))$, we apply the Vinogradov-Korobov bound for $L(s,\chi)$ in order to finalize the $M = \log^{O(1)}(q\,|t|)$, and thus,

$$\begin{aligned}
-\frac{L'}{L}(s,\chi) &= -\sum_{\rho:|\rho-s|\leqslant c_1\delta(t)} \frac{1}{s-\rho} + O\left(\frac{\log M}{\delta(t)}\right) \\
&= -\sum_{\rho:|\rho-s|\leqslant c_1\delta(t)} \frac{1}{s-\rho} + O\left(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\right).
\end{aligned}$$

☐

Considering the quantity $-\Re\frac{L'}{L}(s,\chi)$,

$$-\Re\frac{L'}{L}(s,\chi) = -\sum_{\rho:|\rho-s|\leqslant\delta(t)}\Re\frac{1}{s-\rho} + O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big)$$
$$= O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big).$$

If we have a zero $\rho = \beta + it \in S := \left\{\sigma + it : \sigma > 1 - c\min\Big(\frac{1}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \frac{1}{\log q}\Big), |t| \geqslant 200\right\}$ such that it has the same imaginary part as $s = \sigma + it$ and $|\sigma - \beta| > \delta(t)$, then the bound of $-\Re\frac{L'}{L}(s,\chi)$ is surmountable.

$$-\Re\frac{L'}{L}(\sigma+it,\chi) \leqslant -\frac{1}{\sigma-\beta} + O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big).$$

Expanding the log-derivative of $L(s,\chi)$ along with the familiar inequality regarding the real parts,

$$\begin{cases} -\Re\frac{L'}{L}(\sigma+it,\chi) = \sum_n \frac{\Lambda(n)}{n^\sigma}\Re(\chi(n)n^{-it}) \\ 3 + 4\Re z + \Re z^2 \geqslant 0 \end{cases},$$
$$\Longrightarrow -3\frac{L'}{L}(\sigma,\chi) - 4\Re\frac{L'}{L}(\sigma+it,\chi) - \Re\frac{L'}{L}(\sigma+2it,\chi^2) \geqslant 0.$$

We estimate the two quantities involving log-derivative, noting that $\Re\frac{L'}{L}(\sigma+it,\chi)$ is already derived.

$$-\frac{L'}{L}(\sigma,\chi) = \sum_n \frac{\Lambda(n)}{n^\sigma} = -\frac{\zeta'}{\zeta}(\sigma) = \frac{1}{\sigma-1} + O(1).$$

If χ_1 is the primitive character that induces χ^2, then by Euler product formula,

$$L(s,\chi^2) = \prod_{p\nmid q}\frac{1}{1-\chi^2(p)p^{-s}} = \prod_{p\nmid q}\frac{1}{1-\chi_1(p)p^{-s}} = L(s,\chi_1)\prod_{p|q}1-\chi_1(p)p^{-s}$$
$$\Longrightarrow \frac{L'}{L}(s,\chi^2) = \frac{L'}{L}(s,\chi_1) + \sum_{p|q}\frac{\chi_1(p)p^{-s}}{1-\chi_1(p)p^{-s}}\log p$$
$$\Longrightarrow \left|\frac{L'}{L}(s,\chi^2) - \frac{L'}{L}(s,\chi_1)\right| \leqslant \sum_{p|q}\frac{p^{-\sigma}}{1-p^{-\sigma}}\log p \leqslant \sum_{p|q}\log p \leqslant \log q.$$

Thus, we are able to claim

$$-\Re\frac{L'}{L}(\sigma+2it,\chi^2) = O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big).$$

Therefore, we obtain

$$\frac{3}{\sigma-1} - \frac{4}{\sigma-\beta} + O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big) \geqslant 0.$$

Letting $\sigma = 1 + C(1-\beta)$ for an absolute constant C,

$$\left(\frac{4}{C+1} - \frac{3}{C}\right)\frac{1}{1-\beta} \leqslant O\Big(\min\left(\log q, \log^{2/3}|t|(\log\log|t|)^{1/3}\right)\Big).$$

The contradiction of hypothesis $\beta \in S$ completes the proof.

7.4.3 P.N.T. in Arithmetic Progressions with an Error Term

We introduce the local integrability of log-derivative.

Lemma 7.14. *From the Corollary of $L(s,\chi)$'s log-derivative, we obtain that $\forall C > \varepsilon > 0, t_0 \in \mathbb{R}$,*

$$\int_{\varepsilon}^{C} \int_{t_0-1}^{t_0+1} \left| \frac{L'}{L}(\sigma + it, \chi) \right| dt d\sigma \ll_{\varepsilon, C} \min\left(\log q, \log^{2/3} |t| (\log\log |t|)^{1/3}\right).$$

Proof. Applying Corollary 7.13, the error term is negligible, and each zero in $[t_0 - 1, t_0 + 1] = D(t_0, O_{\varepsilon,C}(1))$ contributes $O_{\varepsilon,C}(1)$ to the integral. The claim follows from the fact that there are $\log\log|t|$ zeros in $\asymp t$ (the proof is trivial). □

In addition, we introduce the truncated Perron's formula.

Lemma 7.15 (Truncated Perron's Formula)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ be such that $\forall n, f(n) = O(\log(2+n))$, and $x, T \geqslant 2$. Then,*

$$\sum_{n \leqslant x} f(n) = \frac{1}{2\pi i} \int_{\sigma - iT}^{\sigma + iT} \mathcal{D}f(s) \frac{x^s}{s} ds + O\left(\left(1 + \frac{x}{T}\right) \log^2 x\right),$$

where $\mathcal{D}f(s)$ is Dirichlet series and $\sigma := 1 + \frac{1}{\log x}$.

Now, we shall tackle the Mangoldt summation.

Theorem 7.16 (Truncated von Mangoldt summation for arithmetic progression)**.** $\forall 0 < \varepsilon < 1; x, T \geqslant 2$,

$$\sum_{\substack{n \leqslant x \\ n \equiv a \pmod q}} \chi(a)\Lambda(n) = x - \sum_{\substack{\rho: \\ \Re\rho \geqslant \varepsilon, \\ \Im\rho \leqslant T}} \frac{x^\rho}{\rho} + O_\varepsilon\left(x^\varepsilon \log^2 T + \frac{x}{T} \log^2 x\right),$$

where ρ represents the zeros of $L(s,\chi)$.

Proof. We claim that $\exists T' \in [T, T+1]$ such that

$$\int_{\varepsilon}^{C} \left| \frac{L'}{L}(\sigma \pm iT', \chi) \right| d\sigma \ll_{C,\varepsilon} \min\left(\log q, \log^{2/3} T (\log\log T)^{1/3}\right)$$

due to the fact that there are t such that $f(t)$ are less than the average of f in $[T, T+1]$, which is the inner integral in Lemma 7.14.

Furthermore, we incorporate the Residue theorem by constructing a set

$$S_0(\varepsilon, T) = \left\{ s \in \mathbb{C} : \varepsilon \leqslant \Re s \leqslant 1 + \frac{1}{\log x}, |\Im s| \leqslant T \right\},$$

and defining $\gamma = \partial S_0(\varepsilon, T')$. For substantial residues $\rho, s = 1$, $\operatorname*{Res}_{s=\rho} \frac{L'}{L}(s,\chi)\frac{x^s}{s} = \frac{x^\rho}{\rho}$, $\operatorname*{Res}_{s=1} \frac{L'}{L}(s,\chi)\frac{x^s}{s} = -x$.

$$\sum_{n \leqslant x} \chi(n)\Lambda(n) = -\frac{1}{2\pi i} \int_{1+\frac{1}{\log x} - iT'}^{1 + \frac{1}{\log x} + iT'} \frac{L'}{L}(s,\chi) \frac{x^s}{s} ds + O\left(\left(1 + \frac{x}{T}\right) \log^2 x\right) \tag{44}$$

$$= -\frac{1}{2\pi i} \int_\gamma \frac{L'}{L}(s,\chi)\frac{x^s}{s} ds + \frac{1}{2\pi i} \int_{1 + \frac{1}{\log x} + iT'}^{\varepsilon + iT'} \frac{L'}{L}(s,\chi) \frac{x^s}{s} ds \tag{45}$$

$$+ \frac{1}{2\pi i} \int_{\varepsilon + iT'}^{\varepsilon - iT'} \frac{L'}{L}(s,\chi)\frac{x^s}{s} ds + \frac{1}{2\pi i} \int_{\varepsilon - iT'}^{1 + \frac{1}{\log x} - iT'} \frac{L'}{L}(s,\chi)\frac{x^s}{s} ds + O\left(\left(1 + \frac{x}{T}\right)\log^2 x\right) \tag{46}$$

From the residue theorem and the observation we made,

$$(46) = x - \sum_{\rho: \rho \in S_0(\varepsilon, T')} \frac{x^\rho}{\rho} - \frac{1}{2\pi i} \int_{\varepsilon - iT'}^{\varepsilon + iT'} \frac{L'}{L}(s,\chi)\frac{x^s}{s} ds + O\left(\min\left(\log q, \log^{2/3} T (\log\log T)^{1/3}\right)\right) + O_\varepsilon\left(\frac{x}{T}\log^2 x\right).$$

In addition, we utilize Lemma 7.14 again to find a ε' that facilitates bounding $\int_{\varepsilon - iT'}^{\varepsilon + iT'} \frac{L'}{L}(s,\chi)\frac{x^s}{s} ds$.

$$\int_{\varepsilon - iT'}^{\varepsilon + iT'} \int_{-T'}^{T'} \left| \frac{L'}{L}(s,\chi)\frac{x^s}{s} \right| dt d\sigma \ll_\varepsilon x^\varepsilon \cdot \min\left(\log q \log T, \log^{5/3} T (\log\log T)^{1/3}\right)$$

By finding a ε' rendering $\int_{-T'}^{T'} \left| \frac{L'}{L}(s,\chi)\frac{x^s}{s} \right| dt$ below its average on $[\varepsilon/2, \varepsilon]$, we obtain the bound of the contour integral.

$$\sum_{n \leqslant x} \chi(n)\Lambda(n) = x - \sum_{\rho : \rho \in S_0(\varepsilon', T')} \frac{x^\rho}{\rho} + O_\varepsilon\Big(x^\varepsilon \min\Big(\log q \log T, \log^{5/3} T (\log\log T)^{1/3}\Big) + \frac{x}{T}\log^2 x \Big) \tag{47}$$

The claim follows from our previous conclusions regarding the contributions of zeros $\notin S_0(\varepsilon', T')$. $\square$

We use $E(x,T)$ to denote the error $O_\varepsilon\Big(x^\varepsilon \min\Big(\log q \log T, \log^{5/3} T (\log\log T)^{1/3}\Big) + \frac{x}{T}\log^2 x \Big)$. The Summation formula we derived is sufficient to transform, by the orthogonality of Dirichlet characters,

$$\begin{aligned}\sum_{\substack{n \leqslant x \\ n \equiv a \pmod q}} \Lambda(n) &= \sum_{n \leqslant x} \left(\frac{1}{\phi(q)} \sum_{\chi \bmod q} \bar{\chi}(a)\chi(n) \right) \Lambda(n) = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \bar{\chi}(a) \sum_{n \leqslant x} \chi(n)\Lambda(n) \\ &= \frac{1}{\phi(q)} \sum_{\chi \bmod q} \bar{\chi}(a) \left(x - \sum_{\rho : \rho \in S_0(\varepsilon', T')} \frac{x^\rho}{\rho} + E(x,T) \right)\end{aligned}$$

due to $(a,q) = 1$.

Coupled with the Zero-Free proposition,

$$\forall \rho = \beta + it,\ \rho \in \left\{ \sigma + it : \sigma \leqslant 1 - c\min\left(\frac{1}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \frac{1}{\log q} \right),\ |t| \geqslant 200 \right\}$$

for Dirichlet $L(s,\chi)$, and

$$x^\beta \leqslant x \exp\left(-c \min\left(\frac{\log x}{(\log T')^{2/3}(\log\log T')^{1/3}}, \frac{\log x}{\log q} \right) \right),$$

we are able to mitigate the growth of

$$\begin{aligned}\sum_{\rho : \rho \in S_0(\varepsilon', T')} \frac{x^\rho}{\rho} &\leqslant \sum_{\rho : \rho \in S_0(\varepsilon', T')} \frac{x^\beta}{|\rho|} \\ &\ll x \exp\left(-\min\left(\frac{\log x}{(\log T')^{2/3}(\log\log T')^{1/3}}, \frac{\log x}{\log q} \right) \right) \log T' \\ &\asymp x \exp\left(-\min\left(\frac{\log x}{(\log T)^{2/3}(\log\log T)^{1/3}}, \frac{\log x}{\log q} \right) \right) \log T\end{aligned}$$

Choosing $\log T \asymp (\log x)^{3/5}(\log\log x)^{-1/5}$ due to $q \leqslant \log^A x$,

$$\sum_{\rho : \rho \in S_0(\varepsilon', T')} \left| \frac{x^\rho}{\rho} \right| \ll_A x \exp\left(-\frac{\log^{3/5} x}{(\log\log x)^{1/5}} \right),$$

which counteracts the increasing rate stemming from $E(x,T)$. Thus, ***Quod Erat Demonstrandum***.

$$\sum_{\substack{n \leqslant x \\ n \equiv a \pmod q}} \Lambda(n) = \frac{x}{\phi(q)} + O\left(x \exp\left(-c_A \frac{\log^{3/5} x}{(\log\log x)^{1/5}} \right) \right). \tag{48}$$

$\square$